

Table 4: **Hpyer-parameters of SFT training for Qwen2.5-7B-Instruct.**

| Hyper-parameter             | Value       |
|-----------------------------|-------------|
| ZeRO Stage (DeepSpeed)      | 3           |
| training epochs             | 3           |
| random seed                 | 42          |
| batch size per device       | 4           |
| gradient accumulation steps | 2           |
| initial learning rate       | 2.e-5       |
| learning rate scheduler     | constant    |
| ratio of warmup steps       | 0.03        |
| weight decay coefficient    | 0.0         |
| adam optimizer              | [0.9, 0.95] |
| adam epsilon                | 1.e-8       |
| use bfloat16                | True        |
| use float16                 | False       |
| max norm of gradient        | 1.0         |
| model max length            | 4096        |

Table 5: **Hpyer-parameters of SFT training for Llama-3.1-8B-Instruct.**

| Hyper-parameter             | Value       |
|-----------------------------|-------------|
| ZeRO Stage (DeepSpeed)      | 3           |
| training epochs             | 3           |
| random seed                 | 42          |
| batch size per device       | 4           |
| gradient accumulation steps | 2           |
| initial learning rate       | 2.e-5       |
| learning rate scheduler     | constant    |
| ratio of warmup steps       | 0.03        |
| weight decay coefficient    | 0.0         |
| adam optimizer              | [0.9, 0.95] |
| adam epsilon                | 1.e-8       |
| use bfloat16                | True        |
| use float16                 | False       |
| max norm of gradient        | 1.0         |
| model max length            | 4096        |

Table 6: **Hyper-parameters of RL training for Gemma-3-12B-IT.**

| Hyper-parameter           | Value |
|---------------------------|-------|
| ZeRO Stage (DeepSpeed)    | 3     |
| micro train batch size    | 2     |
| global train batch size   | 64    |
| micro rollout batch size  | 8     |
| global rollout batch size | 512   |
| max epochs                | 1     |
| number of episodes        | 50    |
| generation max length     | 8192  |
| use bfloat16              | True  |
| actor learning rate       | 2.e-7 |
| critic learning rate      | 1.e-5 |
| initial KL coefficient    | 1.e-3 |
| normalize reward          | True  |
| packing samples           | True  |